

Project part B FINS 3645

Z5642093

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Streamlit Link

GitHub repository: https://github.com/nan336/z5642093_ProjectB

Streamlit app: <https://z5642093projectb-h9husu2hje7ulq6yymvtdu.streamlit.app>

Branch: main

Entrypoint: streamlit_app.py

Funds and Backtest Design

The part A data's basic structure is transformed into six systematically managed funds through Compass Invest's part B's pipeline, which consists of three asset universes and two optimization methods each: an Equity-only universe (50 stocks across 10 sectors), a combined equity-plus-crypto (60 assets), and a Crypto-only universe. In each universe, funds are created using two methods: minimum variance (weights selected in order to minimize historical portfolio volatility, which is solved with a long-only numerical optimizer so no fund can have a negative position in any asset) and equal weight (an equal split over all available assets). To begin with, a walk-forward technique is implemented to help backtest each fund out-of-sample: weights are adjusted on the initial trading day of each month using solely return data that is strictly available prior to that date. Every fund accumulates a six month (126 trading day) preliminary estimation window prior to going "live", which means that the reported performance always represents genuine out-of-sample selections rather than an in-sample fit. Given this time frame, the funds' initial live rebalance date is August 3, 2020, and each fund is assessed throughout a period of 41 monthly rebalances until the sample's conclusion (December 29, 2023). According to the specified rule in the project brief, a risk-free rate of zero is assumed throughout, and transaction expenses are estimated to be zero. To add on, since crypto returns were matched to stock trading days as a component of the Part A data foundation, all fund returns, which includes the Crypto-Only funds, are calculated based on the equity trading calendar instead of the native seven days a week calendar. For internal consistency over all six funds, every annualization in this document utilizes 252 trading days yearly, including Crypto-only funds.

Out Of Sample Results and Fund Fact Sheets

The out-of-sample results for all six base funds across the backtest period of (August 2020 - December 2023) are shown in the (performance_metrics.csv), while (fund_growth_comparison.png) focuses precisely on the combined fund methods and also the sentiment-tilted extension. The results did reveal a unique and consistent pattern: in all of the three asset class universes, equal-weight consistently produces a higher raw return than minimum-variance, while minimum variance consistently yields lower volatility and a lower maximum drawdown, which is exactly the trade off that the two models are intended to produce.

Furthermore, the Crypto only universe, where the two methods differ most from each other, is the best illustration of this trade off. An individual who is thinking of investing in the Crypto-Only Equal Weight fund would have experienced the biggest drawdown in the sample: the fund dropped by 84.8% from its previous peak at the worst point, despite generating a 36.7% annualized return with 73.6% annualized volatility. This shows and suggests that careful and disciplined rule following optimization can benefit and improve risk and return even when there's only a single high volatility asset class.

On the other hand, the dataset tells a different perspective in the Equity-Only universe. In this scenario, Equal Weight (18.1% return, 16.3% volatility, Sharpe of 1.11) surpasses the combined fund consisting of the identical stock's assets and cryptocurrency in terms of risk-adjusted performance. For Compass Invest this really matters throughout this specific sample duration, adding a cryptocurrency diversification to an equity portfolio increases both risk and return, but the risk increased rather more, which made the Sharpe ratio for the Combined fund (1.00) lower than for the Equity-Only alone (1.11). From this info, we can dictate that more diversification is not always beneficial, and it does not directly mean higher risk adjusted returns. This detail is worth noting to a young inexperienced Compass Invest because most novice investors make the mistake of making the assumption or judgement that diversification is always certainly advantageous. Nevertheless, having a little exposure to cryptocurrencies is certainly not useless or worthless; any investors seeking for an increase in absolute returns and also willing to tolerate and deal with greater volatility would choose to pick Combined or Crypto only funds.

Moreover, I looked at the Combined Equal Weight fund's drawdown (drawdown_equal_weight.png), this data shows an actual and true representation of the risk: the fund failed to fully recover to a new all-time high for much of the following year, even though it finished with the sample 78% above its original starting value. The worst of it hit around September 2022, which is around the crypto bear market we discussed in the report paper in **part A**. (growth_of_dollar_sample.png) where ETH-USD crashed from 36 dollars to 8 dollars across the year 2022. As more past data accumulates, the portfolio -weights-over-time (weights_over_time_min_variance.png) highlights how the Combined Minimum-Variance fund gradually distributes its allocation across a greater variety of defensive names (for example, GILD, NEM, and WMT) rather than heavily specializing in a small number of holdings throughout. Furthermore, the risk adjusted performance of all the seven funds, the six base funds from the three assets universes and sentiment-tilted extension covered later in this

document – is summarised in a single view in, (sharpe_by_fund.png). It indicates that Sentiment Tilted fund and Equity-Only Equal Weight hold the two largest sharpe ratio, whereas Crypto-Only Equal Weight has the lowest risk adjusted ranking across the seven funds, given its previous record of high absolute return. Additionally, Crypto-only Minimum Variance represents how careful optimization can help improve this representation even in a risky assets class, with a mid-table Sharpe of 0.77.

The Sentiment Index

In addition, sentiment is analyzed using something known as VADER (Valence Aware Dictionary and Sentiment Reasoner) , which is a lexicon-based sentiment analyzer mostly used for informal or short languages like headlines. VADER is applied to the daily headline panel described in part A, which already gathers all the headlines issued for a certain ticker on a given market trading day into a single compiled string. To remain consistent with VADER's own design-based assumptions, the underlying headline text was maintained in its original form in the entirety of Part A. stop words , punctuation, and casing were intentionally maintained rather than stripped because VADER specifically uses cues – such as exclamation or capitalization- to help identify emphasis and intensity.

Building on this, once VADER is applied, each ticker's day-combined headline text is applied to generate a single "compound" sentiment score, a value that ranges from -1 (most negative) to +1 (most positive). However, since not every ticker trades news daily, the ticker-level sentiment data were reorganized into a full equities trading calendar of 1,006 trading days just before any further processing. This organization process clearly pointed out missing values instead of passively skipping days when a ticker had no headlines. These gaps were then forward filled: a ticker's sentiment score carries forward from its most recent headlines until new news becomes available. This decision was made because, rather than returning to a neutral baseline as soon as coverage stops, market attitude in a firm plausibly endures in between headlines. Another possible approach - treat no-news days as neutral, with a score of zero - was taken into consideration but ultimately rejected because a score of zero could be mistaken for genuinely neutral sentiment (a headline containing both negative and positive wording) rather than the actual insufficient information. The sentiment signal is lagged by just one trading day in relation to the date it is used in order to prevent a look-ahead bias. This makes sure that a fund rebalancing on any given date depends solely on sentiment that was fully known by the end of

the trading day, not same day or further information. A ticker-day's sentiment score is only usable as an input beginning on the following trading day. Prior to being applied to the whole panel, this lag was checked manually on a sample ticker. For instance, a sentiment score calculated for a specific day precisely showed up as the "lagged" value on the row for the very next trading day. This trend continued across gap days as well because the forward-filled value from the previous available headline day carried throughout the shift.

Sentiment Fusion Before vs After

A basic multiplication tilt was also implemented to incorporate the equity Sentiment Index into the Combined Equal Weight Fund: on each rebalance date, the base weight of each equity ticker has been scaled by a factor of $(1 + \text{tilt_strength} \times \text{sentiment})$, with tilt_strength adjusted to 0.5. A neutral tilt factor of 1, which is unchanged, is given to crypto tickers that receive no news coverage. After that, all weights are then renormalized so they sum back to equal one. Moving on, (fusion_comparison.csv) and (fusion_before_after.png) compare and show the base Combined Equal Weight fund against its sentiment-tilted version. The effect is modest but it shows positive results. Sharpe ratio increases from 0.998 to 1.035, max drawdown improves slightly from -28.75% to -28.06%. On the other hand, annualized volatility decreases from 21.07% to 20.30% and annualized returns remain slightly unchanged at 21.02% vs 21.01%. To put it simply, the sentiment tilt had no major effect on the fund's earnings, but it did slightly improve the return's efficiency per unit of risk assumed; this is a truthful, moderate, and evidence-supported change rather than an extraordinary outperformance claim. To add on, there is also one effect of the renormalization step that is worth pointing out: whenever aggregate equity sentiment is strongly negative or positive overall, the entire allocation of cryptocurrency also moves slightly, purely as an effect of renormalization rather than because of any crypto-specific signals. This is not a mistake that needs to be covered up, but rather just a simplification that is allowed for this baseline fusion approach; an improved implementation could precisely help maintain the weight share of the cryptocurrencies, dispersing sentiment-driven changes solely within the equity component.

Innovation and Data-Driven Results

The main extension in this project is not simply the construction of the required Combined fund, but the broader fund menu and the structured comparison across asset universes. Instead of only building the minimum combined equity-plus-crypto fund, Compass Invest offers six

systematic funds: Equity-Only, Crypto-Only, and Combined universes, each implemented with Equal Weight and Minimum-Variance methods. This allows the app to separate three different investor questions: whether crypto improves the equity portfolio, whether optimization reduces risk, and whether high-return crypto exposure is worth the drawdown risk. The results indicated that it yields valuable product knowledge. The highest Sharpe ratio among the six base funds was Equity-Only Equal weight (1.11), while the highest raw return among the equal funds was Crypto-Only Equal Weight (36.7%) with a max drawdown of (84.8%). This is because the application is not just about the highest-returning funds, but rather, the best fund for a novice investor is probably the simpler equity fund.

The App and Investor Journey

Compass Invests fund research is made available using a Streamlit web app designed to help and follow the decision-making of a young novice investor. In the app, the funds tab can be used for comparison and selection of funds: any of the 6 systematic funds can be used or selected, and the user can also see the fact sheet of the fund, such as the annualized return, Sharpe ratio, volatility, growth of \$1, maximum drawdown, and the 10 largest holdings of the funds. The allocation tab also helps to put comparison into the investment decisions. The app does not request the user to manually calculate the trade-off between the portfolio's various funds but allows the user to select the amount of weight to give each fund, re-normalizes the allocation to 100%, and shows the growth path of the blended portfolio as well as its risk signals. Other than that, the Sentiment tab shows and provides the user with a supportive layer of the market, as it is able to display the lagged sector sentiment index created in section 3, which enables comparison between the changes in news tone sectors like Tech, Healthcare, and Energy. The data tab offers a natural language description of the data set, so that the users know what the backtests are derived from. The app purposefully reads precomputed CSV-based artifacts from results/ instead of recomputing backtests or re-scoring the VADER sentiment live. This can be beneficial for the app as it makes it fast and lightweight enough to be used with the free tier of Streamlit Community Cloud, and it aligns with the brief of the project of loading precomputed results in the deployed app.

Reflection

In this particular project, a number of design choices functioned as intended. Instead of providing an exaggerated in-sample fit, the walk-forward, out-of-sample backtest avoided look-ahead bias throughout, producing an accurate representation of each fund's actual performance for an investor. During redevelopment, one specific refinement made is worth mentioning: the historical window originally used to compute weights at each rebalance date was found to include that date's own return, which was inconsistent with the intended "strictly past data only" rule. I fixed this specifically to exclude that date, and it was validated that the equal-weight fund's findings were kept unchanged while the minimum-variance numbers were only slightly modified. This gave indication that the results had not been substantially impacted from the initial implementation, but it was still worth adjusting for methodological precision. In addition, the results themselves provided Compass Invest with a truly valuable outcome: Equity-Only Equal Weight, the simplest and straightforward fund in the list outperformed all other tested funds in terms of risk-adjusted performance (with a Sharpe of 1.11), which shows and remind us that sophisticated investment does not always reflect better results. Folding lagged sentiment into fund weights raised the Sharpe ratio from 0.998 to 1.035 and also marginally decreased both volatility and maximum drawdown without negatively reducing return, which shows the modest but real impact of sentiment fusion.

There are also certain limitations worth taking note of. Firstly, because the sentiment model is built on VADER, which is not really specialized to finance, it may mislead and misinterpret headlines that have terms that have different meanings in financial markets than in everyday speech. Furthermore, the sentiment fusion approach is pretty simple: it renormalizes the portfolio after shifting equity weights, allowing crypto weights to change in the absence of a direct sentiment signal. Thirdly, an inexperienced investor may not notice how unsuitable the cryptocurrency funds are if there's no strong risk control in place, since they combine big drawdowns with substantial rewards.

Recommendations

The first recommendation I make is to use a tool designed specifically for finance sentiment analysis instead of using the generic sentiment tool. VADER's lexicon was designed for the use of social media text in general, and this project's own findings illustrated drawbacks: sentiment remained consistently positive in all sectors for the full 4-year sample throughout the clear market downturns. An immediate next step would be to add more financial-specific terms to VADER's list and their corresponding sentiment scores, as in some instances these words have a clear negative sentiment in the financial domain that may be underrepresented in a general domain lexicon, for example "downgrade," "miss," or "recall" are all words that have a clear negative sentiment in a financial context and may not be given the same negative score by a general domain lexicon as their financial sentiment counterpart. This would likely create a more realistic sentiment signal that would then enable the fusion mechanism in Section 4 to be more successful in extracting a stronger, more distinct signal than the one that the current implementation yields.

Moving on, my second suggestion is to improve the sentiment fusion method so that the Crypto's total weight does not fluctuate as an unintended side effect of renormalization. Although it has no sentiment signal of its own, the current implementation tilts equity weight on the basis of sentiment and then renormalizes the whole portfolio so that the sum equals one. This implies that depending on how positive or negative aggregate equity sentiment occurs on a given rebalance date, cryptocurrency still ends up with a slightly different total allocation. The sentiment-driven renormalization would only be executed inside the equity portion of the portfolio, keeping the equity-versus-crypto split unaffected by the tilt, and a cleaner design would better maintain the original weight share of the cryptocurrency. This would allow for a cleaner, more independent test to determine if sentiment actually increases value, free from the confounding effect of an unintentional crypto-weight side effect, as well as making sure the fusion process effect is easier to explain to a user: "Sentiment helps adjust which stocks that you hold more, not how much crypto you hold".

Furthermore, my third recommendation is to implement specific risk management in place before making crypto-heavy funds available to Compass Invest's public audience who uses the

services offered. There is an obvious need to see the outcome of this project: the Crypto-Only Equal Weight fund had an annualised return of 36.7%, but it also had a maximum drawdown of 84.8%. This means , at the worst time , an investor would have seen the fund decrease to approximately 15% of its previous former high. If an inexperienced young investor is not really familiar with how this works and doesn't understand what is the meaning of an 84.8% drawdown , it might be dangerous to show and present these funds without some security or context. To avoid this from occurring , I recommend capping the allocation for funds with high volatility, implementing a volatility overlay which helps to reduce volatility during periods of high volatility, or alerting in-app users before they confirm the allocation of funds with high volatility where the expected drawdown is over a particular threshold. These controls would help turn Compass Invest's promise of lowering risk into reality , not just a claim or a marketing advertisement.

All things considered, this project demonstrates Compass Invest's core promise to its clients, that is to allow its customers to pick a fund that corresponds to their own risk tolerance. This is not a marketing scheme but it's a promise that is made with statistics that is actually validated. The six-fund menu spans three asset universes, offering real, genuine answers that an inexperienced or new investor may ask: Is there an advantage to add crypto? Does disciplined optimisation work more than spreading money equally? Is it worth gambling on higher returns from crypto? The reflection part of the report also recognizes and solves an authentic methodological issue throughout the process, displaying the same rigor that the application itself is designed to provide for its customers. Compass Invest would be nearer to delivering its promise if the three recommendations mentioned above were implemented accordingly helping a novice investor make a well-informed decision.

Appendix

Tables

1. Performance_metrics.csv

Fund	Return	Volatility	Sharpe	Max Drawdown
Combined Equal Weight	21.02%	21.07%	1.00	-28.75%
Combined Min Variance	7.01%	13.48%	0.52	-18.51%
Equity-Only Equal Weight	18.10%	16.28%	1.11	-20.32%
Equity-only Min Variance	7.33%	13.49%	0.54	-17.91%
Crypto-Only Equal Weight	36.74%	73.55%	0.50	-84.85%
Crypto-Only Min Variance	47.23%	61.29%	0.77	-73.66%

Annualised_return, Annualised_volatility, Sharpe ratio, and maximum drawdown for all six base funds (Aug 2020 - Dec 2023).

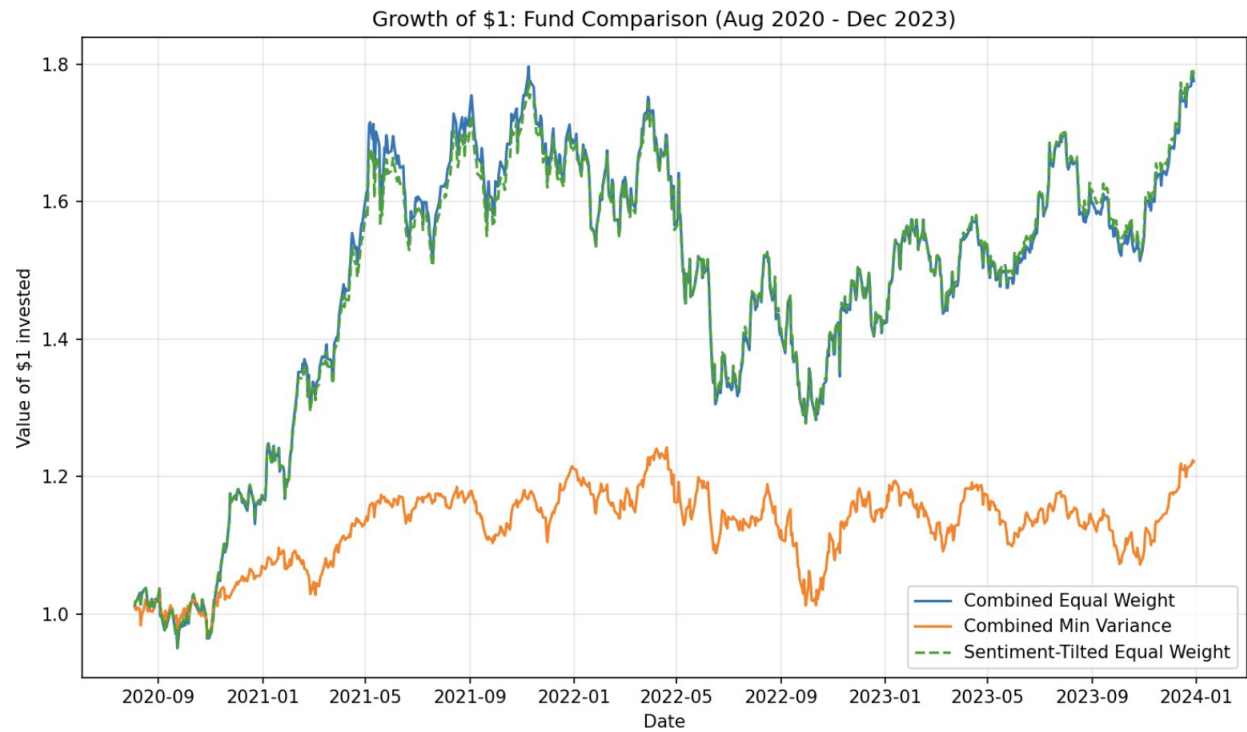
2. fusion_comparison.csv

Fund	Return	Volatility	Sharpe	Max Drawdown
Base (Equal Weight)	21.02%	21.07%	1.00	-28.75%
Sentiment-Tilted	21.01%	20.30%	1.04	-28.06%

Performance comparison of the base Combined Equal Weight fund versus its sentiment-tilted version (Aug 2020 - Dec 2023).

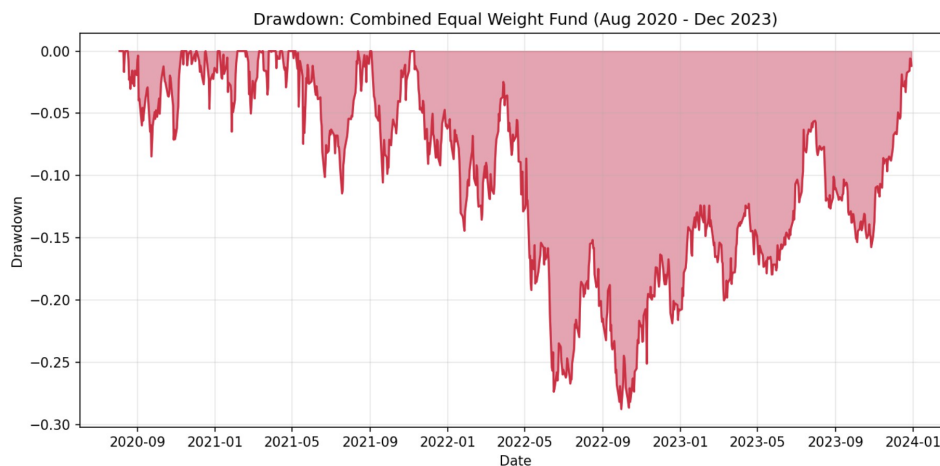
Figures

3. fund_growth_comparison.png



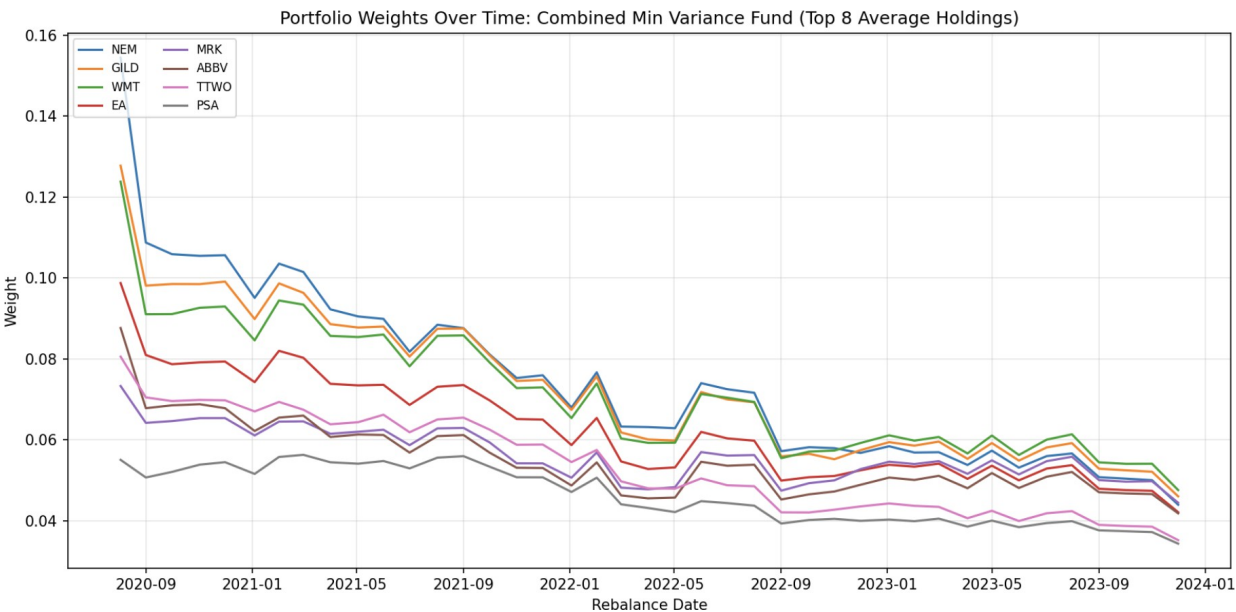
Growth of \$1 for the Combined Equal Weight, Combined Minimum-Variance, and Sentiment-Tilted Equal Weight funds (Aug 2020 - Dec 2023).

4. drawdown_equal_weight.png



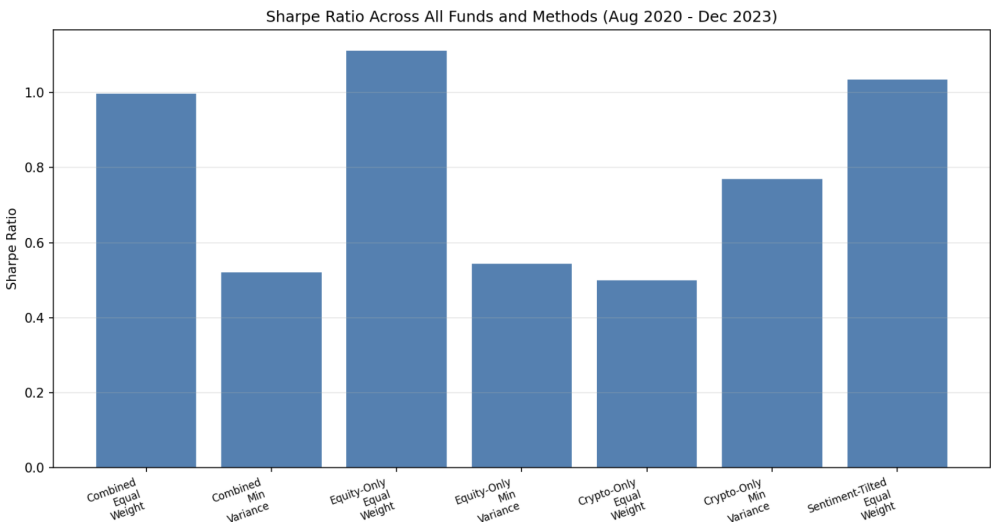
Drawdown over time for the Combined Equal Weight fund (Aug 2020 - Dec 2023).

5. `Weights_over_time_min_variance.png`



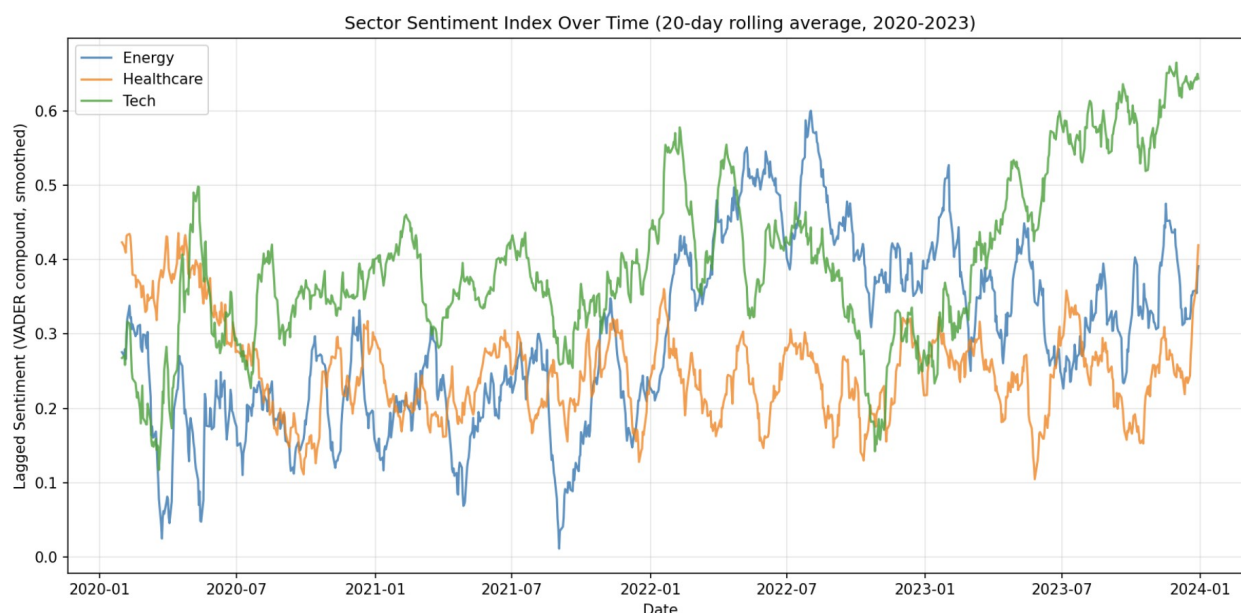
Portfolio weights over time for the top 8 average holdings of the Combined Minimum-Variance fund (Aug 2020 - Dec 2023).

6. `sharpe_by_fund.png`



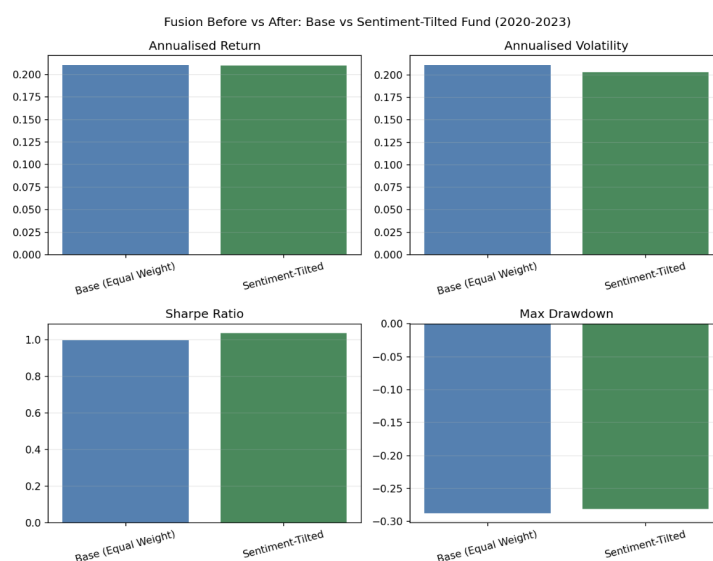
Sharpe ratio across all seven funds - the six base funds plus the sentiment-tilted extension (Aug 2020 - Dec 2023).

7. sector_sentiment_over_time.png



Sector sentiment index over time (20-day rolling average of lagged VADER compound scores) for Energy, Healthcare, and Tech (2020-2023).

8. fusion_before_after.png



Base fund versus sentiment-tilted fund, compared across annualised return, volatility, Sharpe ratio, and maximum drawdown (Aug 2020 - Dec 2023).

References

FINS3645 Course Team (2026) *FinTech Project 2026 dataset: equity_prices, crypto_prices, news_headlines* [Dataset]. UNSW Sydney.

Hutto, C.J. and Gilbert, E. (2014) 'VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text', *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1), pp. 216-225.

Araci, D. (2019) 'FinBERT: Financial Sentiment Analysis with Pre-trained Language Models'. *arXiv preprint* arXiv:1908.10063.

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